INCH-POUND

MIL-P-70506A (AR)
AMENDMENT 2
02 August 1993
SUPERSEDING
AMENDMENT 1
3 April 1989

MILITARY SPECIFICATION

PROJECTILE, 155mm: XM864 BASE BURNER ASSEMBLY
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-P-70506A (AR) dated 4 April 1988, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Change XM864 to "M864" wherever it appears in the specification.

PAGE 3

Add the following new paragraphs as follows:

"2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI Z1.1	- Guide for Quality Control Charts
ANSI Z1.2	- Control Chart Method of Analyzing Data
ANSI Z1.3	- Control Chart Method of Controlling Quality During Production

(Applications for copies should be addressed to American National Standards Institute, 1430 Broadway, N.Y., N.Y. 10018)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

* 3.2.1.1, line 4, Delete: ".441" and substitute: ".514".

3.3.1 Change the paragraph to read as follows:

"3.3.1 Loose igniter composition. After consolidation there shall be no visual evidence of loose igniter composition."

* 3.3.2 Delete "1450+/-50" and substitute "1150+/-50".

PAGE 5

* 3.8 Add new sentence to beginning of paragraph:

"The requirements for the workmanship are as specified on the applicable drawings and referenced specifications."

Delete the second sentence and substitute the following:

"They shall be free of burrs, sharp edges, cracks, gouges, cuts, dirt, grease, rust, and other foreign matter, unless defined elsewhere in this specification."

3.8.1 Change title to read:

"3.8.1 Tracer cups."

3.8.2 Change the paragraph to read:

"3.8.2 Igniter cup, loading assembly. The stepped area of the finished igniter cup need not be perfect form. Damaged areas up to 25 percent of the total surface area are allowed. Damage in excess of 25 percent is cause for rejection."

Add the following new paragraphs:

"3.9 Weather seal assembly. The weather seal assembly shall not allow moisture to leak into the base burner assembly.

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3.9.1 Weather seal resistance to air pressure. The weather seal assembly shall withstand an air pressure test of 15 psi for 15 seconds.

3.9.2 Weather seal shelf life in storage. The shelf life for production weather seals in storage shall be a minimum of one year from the date of their production.

3.9.3 Weather seal degradation. Prior to their use, if it is detected by the base burner manufacturer that a shipment of weather seal assemblies or parts thereof has been contaminated due to exposure to a foreign substance or by the degradation of the packaging assembly, those weather seal assemblies will be rejected and will not be permitted for use.

NOTE: A foreign substance is considered to be any unknown substance; ie. liquid, powder, gel, residue, etc. that does not conform to the provisions of the weather seal assembly drawing, or visually compare in quality , or physical condition to prior shipments."

PAGE 7

4.3.1.1 Examination or test, delete:

"Igniter
(Dwg. 12630572)" and substitute:

"Igniter Cup, Loading Assembly
(Dwg. 12630572)"

PAGE 9

4.4.1 Add the following requirement:

"f. Weather seal assemblies from one lot from one manufacturer."

Add the following new paragraph:

"4.4.1.1 Weather seal assembly raw materials. The raw materials sued in the construction of the M864 w4eather seal assembly will be subject to the following requirements:

a. Aluminum from one lot from one manufacturer.

b. Polyvinyl fluoride (disc seal) from one lot from one manufacturer.

c. Adhesive from one lot from one manufacturer.

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PAGE 9

* 4.4.1 Add "g. Igniter housing plugs from one lot interfix number from one manufacturer."

PAGE 10

4.4.2 Add the following new paragraph:

"4.4.2.c Statistical process control (SPC). For those dimensions identified in the Classification of Defects Tables by the reference "(see Appendix A)", the contractor must utilize the principles of SPC as defined herein, ANSI Z1.1, ANSI Z1.2, ANSI Z1.3, and in Appendix A. When the contractor satisfies the control conditions identified in appendix A, acceptance will then be by SPC. The SPC sample size and frequency of selection shall be determined by the contractor. If the SPC sample is to be less than 4 items per hour, then the approval of the procuring agency must be sought prior to the elimination of sampling inspection for acceptance. The contractor may request to implement SPC acceptance for any other feature that is not annotated in the classification for defects by requesting approval of the procurement agency and by meeting all the conditions specified in this specification. When not accepting product by SPC, acceptance of product for those features identified with references to appendix A will only be in accordance with MIL-STD-414 or MIL-STD-1235 plans CSP-1, -2, -T using variable measurement capable gages."

PAGE 11

4.4.2.2 Change critical defect to read:

"Critical			
Evidence of loose igniter composition	100%	3.3.1	Visual/ Note 2"

Add "(see Appendix A)" after major defects 105, 106 and 107.

Change note 1 to read:

"1. Damaged or missing cadmium plating is permissible on tracer cup provided that the base metal is not gouged or cracked. Flakes of cadmium plating on surface of igniter material is permissible."

Add to the bottom of paragraph 4.4.2.2 under Notes:

* "2. As a referee test for possible loose igniter composition, refer to paragraph 6.8."

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PAGE 12

4.4.2.3 Make the following changes:

Delete Major 101 in its entirety.

Delete Major 102 and substitute the following:

"Major 101	Plug missing	100%	3.1	Visual"

Add:

"Special a	Igniter cup missing or incorrectly oriented (Note 1)	100%	3.1	Visual"

PAGE 13

4.4.2.4 Delete Major 102 in its entirety. Add the following:

"Special a	Igniter cup missing or incorrectly oriented (Note 1)	100%	3.1	Visual"

PAGE 15

* 4.4.2.4 Change drawing "12561476" to "12561471".

PAGE 17

* 4.4.2.8, Category, Major 101, Delete: "or punctured"

Major 102, Delete in its entirety and Substitute:

"102	Interior seal perforation missing or improperly located	100%	3.1	Visual"

Major 104, Delete in its entirety.

PAGE 18

* 4.4.2.9, Special category, Delete special defect d. in its entirety and replace with:

"Special d	Presence of sealant around weather seal base closure interface, 360 degrees	100%	3.1	Visual"

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Major category, Add:

"103	Interior seal perforation not covered with sealant	0.40%	3.1	Visual
104	Exterior weather seal missing or punctured	100%	3.1	Visual
105	Exterior weather seal improperly positioned	0.40%	3.1	Gage"

Delete minors 201, 202, 203 in their entirety and substitute the following:

"Minor				
201	M864 BASE BURNER ASSEMBLY notation and lot number missing, misleading or unidentifiable	.65%	3.1	Visual
202	Propelling grain batch number missing, or unidentifiable	.65%	3.1	Visual
203	Evidence of poor workmanship	.65%	3.8	Visual

* 4.4.2.10 Add new paragraph (attached)

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Quality conformance inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70506A (AR)
AMENDMENT 2

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.10	Packing and Marking for Base Burner Assembly for M864 Projectile	SHEET 1 OF 1	12561471
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Improperly packed	.65%	3.1	Visual
202	Marking missing, incorrect or illegible	.65%	3.1	Visual
203	Evidence of poor workmanship	.65%	3.8	Visual

NOTES:

7 (18a)

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PAGE 19

* Delete 5.2.1 in its entirety and substitute the following:

"5.2.1 Domestic Shipments. Packaging, packing, and marking of the base assembly shall be in accordance with 12561471.

PAGE 20

* 4.5.2 Delete in its entirety and substitute the following:

"4.5.2 Heat of explosion, igniter composition. The igniter material shall be tested for heat of explosion (combustion) in a closed bomb calorimeter in accordance with 4.5.11. Two (2) 3 gram samples shall be taken from each batch of igniter composition produced. If both samples fail to yield a calorie per gram value in the specified range, (see 3.3.2) reject the material batch. If one sample fails, take a second double sample. If one or both samples fail from the second double sample, then reject the batch. Accept a batch only when both samples of a double sample fall within the specified range."

* 4.5.4 Delete "(25 Atmospheres Nitrogen)" and substitute "(25 Atmospheres inert gas)"

PAGE 22

* 4.5.8, line 7, Delete: "M557 point detonating fuze" and Substitute: "M577/M577A1 fuze set for point detonation"

PAGE 23

* 4.5.8, line 3, after "3.2.", Add: "If any lot fails to meet the RPE requirement of 3.2.1.1 (as calculated per 4.5.8.3), the lot shall be referred to the technical agency for disposition."

PAGE 24

* 4.5.10 Add the following new paragraph:

"4.5.10 HEAT OF EXPLOSION OF IGNITER COMPOSITION USING THE PARR OXYGEN BOMB ADIABATIC CALORIMETER

a. PRINCIPLE AND STATUS

The heat of explosion of igniter composition material is determined by ignition of a known mass in an argon atmosphere in a Parr Adiabatic Calorimeter.

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b. APPARATUS AND REAGENTS

1. Calorimeter, Parr Oxygen Bomb, Adiabatic, Automatic

2. Balance, Analytical, 0.1 mg.

3. Balance, Torsion, 0.1g, 4kg. capacity

4. Scissors

5. Wash bottle

6. Water heater

7. Water cooler

8. Argon gas

9. Water, distilled

c. 1. Preliminary

Determine the water equivalent factor of the calorimeter.

2. Sample size and Bomb Pressure

Sample Size	Bomb Pressure
Two (2) 3 gram samples of igniter composition from each batch	25 atm of inert gas (Argon)

d. PROCEDURE

1. Fill the calorimeter bucket by first taring the dry bucket on a solution or trip balance; then add 2000 (+/- 0.5) grams of water. Distilled water is preferred, but demineralized or tap water containing less than 250 ppm of dissolved solids is satisfactory. The water temperature should be approximately 1.5 C below room temperature, but this can be varied to suit the operator's preference. It is not necessary to use exactly 2000 grams, but the amount selected must be duplicated within +/- 0.5 gram for each run.

2. Closing the bomb. Care must be taken not to disturb the sample when moving the bomb head from the support stand to the bomb cylinder. Be sure that the contact ring is in place above the sealing ring and that the sealing ring is in good condition; then slide the head into the cylinder and push it down as far as it will go. It will be convenient (but not essential) to hold the bomb in a bench clamp during the closing and while filling the bomb with argon. Set the screw cap on the cylinder and turn it down firmly by hand. Do not use a wrench or spanner on the cap. Hand tightening should be sufficient to secure a tight seal.

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3. Filling the bomb. Press the fitting on the end of the hose into the inlet valve socket and turn the knurled union nut finger tight. Close the valve on the filling connection; then open or "crack" the argon tank valve not more than one-quarter turn. Open the connection control valve slowly and watch the gage as the bomb pressure rises to the desired filling pressure (usually 30 atm., but never more than 40 atm.); then close the control valve. The bomb inlet check valve will close automatically when the supply is off, leaving the bomb filled to the highest pressure indicated on the atm. gage. Fill the bomb to 30 atm. and exhaust a total of 3 times before filling to the appropriate pressure for the particular sample being tested. When testing a powder or granular material, the filling and purging operation must be done very slowly so the sample will not be disturbed. Release the residual pressure in the filling hose by pushing downward on the lever attached to the release valve. The gage should now return to zero. If the pressure drops slowly and a large amount of gas escapes when the pressure relief valve is opened, the check valve in the bomb head is not operating properly. This trouble will have to be corrected before the bomb can be used. If too much should accidentally be introduced into the bomb, do not proceed with the combustion. Detach the filling connection; exhaust the bomb; remove the head and reweigh the sample before repeating the filling operation.

4. Set the bucket in the calorimeter; attach the lifting handle to the two holes in the side of the screw cap and lower the bomb into the water with its feet spanning the circular boss in the bottom of the bucket. Handle the bomb carefully during this operation so that the sample will not be disturbed. Remove the handle and shake away any drops of water back into the bucket; then push the two ignition lead wires into the terminal sockets on the bomb head, being careful not to remove any water from the bucket with the fingers.

5. Close the calorimeter cover; lower the thermometer bracket; adjust the reading lens and turn on the power switch to start the motor.

6. Let the calorimeter run for 7 minutes while the controller brings the jacket temperature up to equilibrium with the bucket. After the temperatures have equalized, vibrate the thermometer and read the bucket temperature to one-tenth of the smallest scale division.

7. Stand back from the calorimeter and fire the bomb by pressing the ignition button and holding it down until the indicator light goes out. Normally the light will glow for only about 1/2 second, but release the button with 5 seconds regardless of the light.

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8. The bucket temperature will start to rise within 20 seconds after firing. The rise will be rapid during the first few minutes; then it will become slower as the temperature approaches a stable maximum. It is not necessary to observe temperatures during the rapid rise period since the controller will make the adjustments needed to keep the bucket and jacket temperatures equal.

9. Start temperature readings at about 7 minutes after firing, vibrating the thermometer to settle the mercury before each observation. Read the thermometer at one minute intervals until the temperature reaches a stable maximum and remains constant for at least two minutes (three successive readings); then read and record this final temperature to one-tenth of the smallest scale division.

10. Raise the thermometers; swing the cover aside and lift the bucket out of the calorimeter, but let the motor continue to run while the calorimeter is open. Move the jacket temperature control to the purge position to lower the temperature to the starting point in preparation for another run.

11. Remove the bomb from the bucket and open the knurled valve knob on the bomb head to release the residual gas pressure before attempting to remove the cap. Venting the bomb must be carried out either in the hood or out of doors. If no further use is to be made of the combustion products, wash the bomb thoroughly and allow to drain. If further analyses are to be made, the bomb washings should be caught and retained for analysis.

Calculations

$$\text{Heat of Explosion, cal/g} = \frac{WE * \text{delta } T}{W}$$

Where WE = Water equivalent factor
delta T = Temperature rise
W = Weight of sample

When Booster propellant is required, the calculation isadjusted as follows:

$$\frac{(WE * \text{delta } T - CWB * C)}{W} =$$

Where WB = Weight of booster
C = Caloric value of booster

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e. SAFETY ASPECTS

Do not exceed calculated maximum sample sizes or fill the bomb to greater than forty (40) atmospheres pressure. Otherwise excessive pressures may develop in the bomb. Exercise all safety precautions normally taken when handling solid propellant and gas cylinders."

* Delete 5.2 in its entirety and substitute the following:

"5.2 Domestic Shipments. Packaging, packing and marking of the base burner assembly shall be in accordance with 12561471."

Add the following new paragraphs:

"5.3 Storage of approved weather seal lots. Upon completion of a pre-determined quantity of production weather seal assemblies, the seal assemblies shall be packed in a government approved storage container to prevent decontamination of the seals. As the seals come off of the production line, they will have a maximum allowable time of six (6) hours dormancy before being required to be packed for storage.

5.3.1 Date of manufacture. The date of manufacture, the weather seal lot number, and the name of the manufacturer for that weather seal lot will be recorded and included with each container of weather seals."

PAGE 25

* 6.8 Change paragraph number to "6.9" and add:

"6.8 Drop test for determining loose igniter composition. If visual inspection is questionable for determining loose igniter composition the following test will be implemented:

Place the subject igniter cup face down (open end down) onto a clean, flat piece of blank, unlined white paper. Manually raise the subject igniter cup face down above the paper and release it, allowing it to free fall onto the paper. The igniter cup should be dropped from a height of one-half inch (.5 inch) approximately by eye, so that the igniter cup strikes face down. If any igniter composition is deposited on the paper, that igniter cup will be rejected. At the start of each new test, a new piece of paper will be used."

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Add new Appendix A, pages (26) through (28).

"APPENDIX A

10. SCOPE

10.1 Scope. This Appendix details the procedure for implementation of statistical process control for articles covered in this specification. This Appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this Appendix.

30. REQUIREMENTS

30.1 Application of statistical process control (SPC). The contractor will for the identified dimensions maintain Xbar and R SPC charts. After the contractor has demonstrated that its process for manufacturing a particular feature is in control, as defined in paragraph A.1.1; it will discontinue acceptance sampling inspection in favor of acceptance by SPC. However, SPC charting will be maintained for all identified dimensions whether or not the process is in control. It is expected that the contractor will make every effort to bring each process into control. It is required that the contractor re-evaluate its established control limits on a monthly basis and make those changes warranted by the data. At no time should its process parameters vary such that it will fail to meet the requirements established in subsequent paragraphs. Should this happen, the contractor will return to acceptance by sampling.

30.1.1 Implementation of SPC for acceptance of product. For those dimensions identified in the classification of defects tables as being applicable for elimination of lot sampling acceptance, the contractor shall implement, on a dimension by dimension basis, acceptance by SPC after it conforms to each of the following steps:

a. The contractor performs an in-depth process capability study to establish the mean process average (Xdouble bar), Rbar, upper and lower control limits. This study will consist of at least 40 subgroups of data collection.

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b. The contractor demonstrates that its process is under control for at least 20 consecutive production days. Control is defined as remaining within the established control parameters and not exhibiting any of the out of control conditions defined in paragraph A.1.2.a through A.1.2.e more than two times on an individual character basis.

c. Thirty days consecutive production have passed acceptance sampling for the feature.

d. The process has demonstrated Cpk equal to or greater than 1.3 as demonstrated when conducting the above test. Cpk is defined as follows:

$$Cpk = \frac{\text{process mean - nearest specification limit}}{3 \text{ sigma}}$$

sigma = process standard deviation (calculated by the range method per the referenced ANSI documents)

e. The contractor documents its detailed process description. This process description must be submitted to Commander, AMCCOM, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07806-5000 for review and disapproval if the description does not meet its objectives. That is the process description will identify all known contributing variables and their respective controls that are necessary to maintain the specific dimension (i.e., weight, temperature, machine speed, time, tool life, solution proportion, etc.). The Government will furnish the contractor with written notice of the acceptability of its process controls. A notice of control acceptability is required prior to the implementation of acceptance by SPC.

30.1.2 Process control analysis. Once SPC acceptance is implemented, the contractor will evaluate its process for evidence of nonrandom variation (e.g. trends, extreme values, etc.). Should any out of control condition be detected, the product produced from the last sample till the process is brought back into control will be isolated for analysis. The contractor will bring the process back into control or shut the process down. Upon determining the cause of the out of control condition, the contractor will determine if the product manufactured and isolated could be out of engineering tolerance. If the out of control condition did not result in the production of nonconforming material, the material will be released. If the out of control condition could have created nonconforming product, the contractor will sample the product using MIL-STD-105 and the AQL specified in the classification of defects table to determine the acceptability of the isolated product. This evaluation of out of control product requires the review and approval of the local Government quality assurance representative. As a minimum, the contractor will look for any of the following out of control indications:

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a. If any point goes outside the Xbar of R chart control-limits.

b. If 9 points in a row are either continually above or continually below the Xdouble bar.

c. If 6 points in a row are either continually increasing or continually decreasing on the Xbar chart.

d. If any individual sample, Xi, falls outside of the specification limits.

30.1.2.1 Process capability review. At least once a month the contractor will review its process data and establish updated control limits.

30.1.3 Mandatory return to sampling acceptance. Acceptance by SPC will be discontinued in accordance with the following paragraphs. In any case, control charting will be continued. When this occurs, the contractor may reintroduce acceptance by SPC by repeating the steps outlined in paragraph A.1.1.

30.1.3.1 Out of control incidents. If for any individual process, an out of control incident is identified 3 times within 5 consecutive shifts or 5 times within 30 consecutive shifts; and the analysis indicated that nonconforming product may have been manufactured, the contractor will return to acceptance by sampling for the pertinent feature.

30.1.3.2 Process interruption. If the process is interrupted for more than 10 calendar days, the contractor will return to acceptance by sampling for that pertinent feature.

30.1.3.3 Loss of control. If either by testing or data review it is determined that the Cpk is no longer at the required minimum of 1.3, the contractor will return to acceptance by sampling for the pertinent feature.

30.1.4 Records. The contractor will maintain all records dealing with SPC in its files for a minimum of 5 years after completion of the contract.

30.2 Government auditing. In order to verify the state of control of the process, it is recommended that Government representatives select samples at unscheduled times. The sample mean and range should then be compared to the data being recorded by the contractor. If the data is not consistent with the reported data, corrective action should be sought and if necessary, screening of material should be required.

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